electric field. The force field around an electric charge, which pulls and pushes on other electric charges.

electric field lines. Lines that point in the direction of the force that an electric field exerts on charged particles. Electric analogue of magnetic field lines.

electromagnetic waves. Waves of electric and magnetic forces. These include, depending on the wavelength, radio waves, microwaves, infrared radiation, light, ultraviolet radiation, X-rays, and gamma rays.

electron. A fundamental particle of matter, with negative electric charge, which populates the outer regions of atoms.

electron degeneracy. The behavior of electrons at high densities, in which they move erratically with high speeds as a result of quantum mechanical wave/ particle duality.

elementary particle. A subatomic particle of matter or antimatter. Among the elementary particles are electrons, protons, neutrons, positrons, antiprotons, and antineutrons.

embedding diagram. A diagram in which one visualizes the curvature of a two-dimensional surface by embedding it in a flat, three-dimensional space.

entropy. A measure of the amount of randomness in large collections of atoms, molecules, and other particles; equal to the logarithm of the number of ways that the particles could be distributed without changing their macroscopic appearance.

equation of state. The manner in which the pressure of matter (or matter's resistance to compression) depends on its density.

equivalence principle. See *principle of equivalence.*

error box. The region of the sky in which observations suggest that a specific star or other object is located. It is called an error box because the larger are the uncertainties (errors) of the observations, the larger will be this region.

escape velocity. The speed with which an object must be launched from the surface of a gravitating body in order for it to escape the body's gravitational pull.

event. A point in spacetime; that is, a location in space at a specific moment of time. Alternatively, something that happens at a point in spacetime, for example, the explosion of a firecracker.